

EX:NO:20

PROGRAM:

```
def insertion_sort(arr):  
    for i in range(1, len(arr)):  
        key = arr[i]  
        j = i - 1  
        while j >= 0 and arr[j] > key:  
            arr[j+1] = arr[j]  
            j -= 1  
        arr[j+1] = key  
    return arr  
  
# Tests  
print(insertion_sort([3,1,4,1,5,9,2,6,5,3]))  
# [1,1,2,3,3,4,5,5,6,9]  
  
print(insertion_sort([5,5,5,5]))  
# [5,5,5,5]  
  
print(insertion_sort([2,3,1,3,2,1,1,3]))  
# [1,1,1,2,2,3,3,3]
```

Output

```
[1, 1, 2, 3, 3, 4, 5, 5, 6, 9]  
[5, 5, 5, 5]  
[1, 1, 1, 2, 2, 3, 3, 3]
```

=== Code Execution Successful ===